

QUALICUM BEACH, BC, Canada

WMO: 717660

Lat: 49.340N

Lon: 124.390W

Elev: 191

StdP: 14.59

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.3	26.0	14.5	11.6	27.5	19.4	14.7	29.6	18.5	42.9	17.0	42.5	3.1	240	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	82.4	63.1	78.6	62.1	75.2	61.2	64.8	76.7	63.7	74.7	62.5	72.4	5.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	80.8	67.4	59.6	76.7	66.6	58.4	73.4	65.9	29.9	77.1	29.0	74.9	28.2	72.6	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	13.4	11.6	16.6	88.3	3.3	3.3	14.2	90.7	12.2	92.7	10.3	94.6	7.9	97.0
			16.3	67.2	3.6	1.3	13.7	68.2	11.6	68.9	9.5	69.7	6.9	70.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	38.3	39.3	42.3	46.7	54.6	59.2	64.6	64.4	57.8	48.8	42.3	37.5	(6)
	DBStd	10.83	5.23	5.04	4.33	4.13	5.16	4.95	4.21	4.14	4.63	4.33	5.68	5.50	(7)
	HDD50	1736	363	299	241	114	15	0	0	0	2	75	238	389	(8)
	HDD65	5688	828	719	704	550	322	188	56	62	221	503	682	853	(9)
	CDD50	1629	0	0	2	14	158	276	454	447	235	37	6	0	(10)
	CDD65	108	0	0	0	0	0	14	45	44	5	0	0	0	(11)
	CDH74	980	0	0	0	0	26	139	395	370	50	0	0	0	(12)
	CDH80	209	0	0	0	0	1	32	93	77	6	0	0	0	(13)
(14) Wind	WSAvg	4.4	4.5	4.5	5.1	4.7	4.2	4.3	4.1	3.7	3.8	4.3	4.8	4.8	(14)
(15) Precipitation	PrecAvg	42.7	6.7	4.0	3.9	2.4	1.8	1.7	0.9	1.3	1.6	4.6	6.6	6.7	(15)
	PrecMax	53.0	12.6	12.8	8.3	5.1	3.5	3.2	1.7	4.6	5.3	10.2	13.3	9.3	(16)
	PrecMin	30.4	3.4	0.2	0.8	0.7	0.5	0.4	0.1	0.1	0.4	0.7	2.2	3.4	(17)
	PrecStd	6.5	2.4	3.0	1.9	1.3	0.8	0.7	0.5	1.1	1.2	2.3	3.3	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	53.0	58.9	66.2	77.3	84.4	87.9	86.2	79.8	64.7	57.4	52.8	(19)
		MCWB	47.7	48.8	50.9	52.9	58.7	62.0	65.7	64.1	62.9	56.5	52.8	49.1	(20)
	2%	DB	49.0	50.3	53.9	60.5	73.0	78.2	82.4	82.1	74.1	61.6	54.8	48.7	(21)
		MCWB	46.6	46.0	47.6	49.3	57.5	61.0	63.7	63.1	60.3	54.3	51.2	46.5	(22)
	5%	DB	47.2	48.3	51.3	57.3	68.6	72.4	78.2	78.2	70.4	58.5	52.5	46.7	(23)
		MCWB	45.2	45.0	46.3	48.7	56.1	59.2	62.3	62.6	59.6	53.2	49.9	44.8	(24)
	10%	DB	45.5	46.7	49.5	54.7	64.8	68.4	74.8	74.4	67.0	56.3	50.3	45.0	(25)
		MCWB	43.7	44.1	45.3	47.7	55.2	57.8	61.2	61.9	58.1	52.1	48.2	43.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.5	50.6	52.3	55.5	60.7	64.5	67.2	66.4	65.1	58.9	54.6	50.2	(27)
		MCDB	50.4	52.1	57.0	61.5	72.2	77.0	82.5	77.9	75.1	63.0	56.3	52.0	(28)
	2%	WB	47.1	47.6	49.1	52.1	58.8	62.2	64.8	64.8	62.7	56.4	52.0	47.0	(29)
		MCDB	48.5	49.0	52.5	57.9	68.9	74.5	77.2	76.5	70.8	59.4	53.7	48.3	(30)
	5%	WB	45.4	45.8	47.5	50.2	57.4	60.4	63.6	63.7	60.8	54.4	50.3	45.0	(31)
		MCDB	46.8	47.3	50.2	54.4	66.5	69.9	74.8	74.8	68.2	57.1	52.0	46.3	(32)
	10%	WB	43.9	44.5	46.1	48.8	55.9	58.8	62.4	62.6	59.2	52.5	48.6	43.5	(33)
		MCDB	45.2	46.1	48.7	52.9	63.3	66.6	72.3	72.2	64.8	55.3	50.2	44.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	10.7	12.4	14.5	16.3	15.5	18.1	18.0	16.0	12.9	10.9	9.0	(35)
		MCDBR	9.9	11.8	14.2	18.9	24.5	23.3	25.3	24.6	23.2	16.2	11.2	9.2	(36)
	5% WB	MCWBR	7.3	8.3	9.1	10.4	11.0	9.0	9.7	9.6	11.4	10.2	8.2	7.4	(37)
		MCWBR	9.1	9.2	12.4	14.9	19.8	18.9	21.2	20.5	18.7	13.5	10.0	8.4	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.315	0.307	0.327	0.340	0.363	0.356	0.346	0.347	0.332	0.332	0.314	0.309	(40)
		taud	2.511	2.581	2.534	2.491	2.443	2.480	2.551	2.554	2.584	2.555	2.544	2.484	(41)
	Ebn at Noon	Ebn at Noon	238	272	283	289	285	286	288	283	278	257	236	222	(42)
		Edh at Noon	19	22	27	31	34	33	30	29	26	23	18	17	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	267	550	820	1274	1650	1782	1947	1661	1177	638	318	210	(44)
	RadStd	RadStd	38	85	108	124	167	195	184	138	101	98	57	31	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page